Reducing endpoint-transition force spikes

Local quadrature transfer for IPC Toolkit issue 280

1. Old formula and the source of the spike

Consider one source element with vertices $x_1, \ldots, x_n$. All point potentials and sample weights below depend on the current positions. The total energy sums the following contributions over source elements.

Variables. $w > 0$ is the existing, fixed rest-geometry prefactor; n is the source vertex count (triangle: 3; rod: 2; particle: 1). N_i and F_i are the existing near and far point potentials at vertex x_i, summed over its target interactions. q_j is an interior edge-edge (EE) sample on a source edge; Q_j is its near point potential. $m_j \geq 0$ is its existing quadrature weight, including distance support and the endpoint-transition mollifier. Set $M = \sum_j m_j$.

$$\boxed{E_{\text{old}} = w \frac{\sum_i N_i + \sum_j m_j Q_j}{n + M} + \frac{w}{n} \sum_i F_i.} \tag{1}$$

Let $A = (\sum_i N_i + \sum_j m_j Q_j)/(n + M)$ be the normalized near average. The part of the gradient caused by changing the EE weights is

$$\nabla E_{\text{old}}|_{\nabla m} = \frac{w}{n + M} \sum_j (Q_j - A) \nabla m_j. \tag{2}$$

Near an endpoint transition, ∇m_j can be large over a narrow interval. Even when q_j nearly duplicates one endpoint, $Q_j - A$ can remain large because A includes other, distant vertices. The weight therefore changes the energy substantially while barely changing the sampled contact location. Physical force is $-\nabla E$.

In the edge-above-box scene, the old horizontal force at the spike is 103.127: weight derivatives contribute 103.152, and direct potential derivatives contribute approximately -0.025. The edge angle is about 24.5°; the relevant transition is at an endpoint.

2. New formula and intuition

For sample j on source edge (a_j, b_j), let $u_j \in (0, 1)$ be its interpolation coordinate. Introduce a *new* endpoint reference R_j:

$$q_j = (1 - u_j) x_{a_j} + u_j x_{b_j}, \qquad R_j = (1 - u_j) N_{a_j} + u_j N_{b_j}, \qquad \Delta_j = Q_j - R_j. \tag{3}$$

With $D = 1 + M$, replace global renormalization by local transfer:

$$\boxed{E_{\text{new}} = \frac{w}{n} \left[\sum_i (N_i + F_i) + \frac{\sum_j m_j \Delta_j}{D} \right].} \tag{4}$$

Start with vertex weights $1/n$. Give q_j weight $\alpha_j = m_j/(nD)$, taken from its own two endpoints in proportions $1 - u_j$ and u_j. The subtraction of R_j removes exactly that donated endpoint contribution. Thus $Q_j - R_j$ measures the change from moving quadrature mass to q_j; it need not be positive. The mollifier itself and the far term are unchanged.

3. Why the new formula has a smaller spike

For a single EE sample, the weight-gradient term becomes

$$\nabla E_{\text{new}}|_{\nabla m} = \frac{w}{n}\frac{Q-R}{(1+m)^2}\nabla m. \tag{5}$$

As the sample reaches its source endpoint, $Q \to N_a$ and $R \to N_a$, provided the limiting point-potential stencils agree. The coefficient of the steep mollifier gradient then vanishes. The old coefficient compares the sample with an element-wide average, which need not match that endpoint. For an affine near-potential field along the edge, $Q = R$ exactly and the correction is zero.

With multiple samples, write $S = \sum_j m_j \Delta_j$. The exact weight-gradient term is

$$\nabla E_{\text{new}}|_{\nabla m} = \frac{w}{nD}\sum_j \left(\Delta_j - \frac{S}{D}\right)\nabla m_j. \tag{6}$$

The shared denominator couples the residuals, so one zero residual does not eliminate every weight force. Target-endpoint transitions also need not match the source reference. This explains reduction rather than guaranteed elimination of every spike. In the native scene sweep, the maximum $|F_{A,x}|$ drops from 103.13 to 8.89 (a 91.38% reduction).

4. Positivity and divergence: what is preserved

Define $b_{ij} = 1 - u_j$ at $i = a_j$, $b_{ij} = u_j$ at $i = b_j$, and zero otherwise. The new near energy is equivalently

$$E_{\text{new,near}} = w\left(\sum_i \beta_i N_i + \sum_j \alpha_j Q_j\right), \qquad \beta_i = \frac{1}{n} - \sum_j \alpha_j b_{ij}. \tag{7}$$

Because $0 \le b_{ij} \le 1$ and $\sum_i b_{ij} = 1$,

$$\alpha_j \ge 0, \qquad \beta_i \ge \frac{1}{nD} > 0, \qquad \sum_i \beta_i + \sum_j \alpha_j = 1. \tag{8}$$

Positivity. This quadrature sum has nonnegative coefficients even though Δ_j can be negative. Consequently, if the aggregate point potentials N_i, Q_j, F_i are nonnegative, then $E_{\text{new}} \ge 0$ (strictly positive when a positive-coefficient potential is positive).

Divergence. Under those same nonnegativity assumptions, the coefficients satisfy

$$\beta_i \ge \frac{1}{n(n+M)}, \qquad \alpha_j \ge \frac{m_j}{n(n+M)} \quad \Longrightarrow \quad E_{\text{new}} \ge \frac{1}{n}E_{\text{old}}. \tag{9}$$

For the summed energy, $n \le 3$ gives $E_{\text{new}}^{\text{total}} \ge E_{\text{old}}^{\text{total}}/3$. Thus any valid divergence guarantee for the old discretization transfers to the new one. On a fixed finite mesh with fixed supports, uniformly lower-bounded aggregate near potentials and a uniformly bounded far contribution give $E_{\text{new}}^{\text{total}} \ge E_{\text{old}}^{\text{total}}/3 - C$, with C independent of the shrinking gap, which also preserves divergence.

Scope of the claim. Positive coefficients are established algebraically. The code's point potentials use signed inclusion–exclusion sums; nonnegative primitive barriers alone do not establish nonnegative aggregates. Aggregate bounds and the old discretization's contact-coverage/divergence guarantee remain assumptions here. This is a conditional preservation argument, not a completed global positivity or collision-barrier proof.